

1. Use Case Description

1.1. System Overview

The Vehicle Tax Management System is designed to help a city manage information about vehicles, their owners, and the corresponding annual taxes. The system supports two types of vehicles: cars and motorcycles. Each vehicle is identified by a unique plate number and contains information about its maker and model. Similarly, each person is identified by a unique license number, and their name, surname, and address are stored in the system.

The system allows users to register, modify, delete, and search for both people and vehicles. It also manages vehicle ownership, including assigning a vehicle to a person and transferring ownership between people.

Additionally, the system calculates annual vehicle taxes based on the vehicle type and generates an annual report containing vehicle owners, their vehicles, individual vehicle taxes, and the total tax amount for each owner.

1.2. System Actor

City Administrator is the primary actor who interacts with the system. The administrator is responsible for managing people and vehicles, updating ownership information, searching for records, and generating annual tax reports.

1.3. Use Case Descriptions

UC1 – Add Person

The City Administrator registers a new person by entering their license number, name, surname, and address. The system checks that the license number is not already registered and stores the new person's information.

UC2 – Edit Person

The City Administrator selects an existing person and modifies their information, such as their name, surname, or address. The system saves the updated information.

UC3 – Delete Person

The City Administrator selects an existing person to remove from the system. The system deletes the corresponding record. If the person owns vehicles, the system must handle their ownership before deletion according to the chosen deletion policy.

UC4 – Search Person

The City Administrator searches for a registered person using their license number or other available information. The system displays the matching records.

UC5 – Add Vehicle

The City Administrator registers a new vehicle by entering its plate number, maker, model, and type. Additional information is provided depending on the vehicle type, such as engine displacement for motorcycles or CO2 emissions for cars. The system checks that the plate number is unique and stores the vehicle.

UC6 – Edit Vehicle

The City Administrator selects an existing vehicle and updates its information. The system saves the changes, which may affect the vehicle's annual tax.

UC7 – Delete Vehicle

The City Administrator selects a vehicle to remove from the system. The system deletes its record and removes the corresponding ownership association, if one exists.

UC8 – Search Vehicle

The City Administrator searches for a vehicle using its plate number or other available information. The system displays the matching vehicles and their details.

UC9 – Assign Vehicle

When an unowned vehicle is sold to a person, the City Administrator selects the vehicle and its new owner. The system verifies that both records exist and assigns the vehicle to the selected person. It then calculates the corresponding annual vehicle tax.

UC10 – Transfer Vehicle

The City Administrator selects a vehicle that belongs to a person and specifies its new owner. The system updates the ownership information by removing the association with the previous owner and assigning the vehicle to the new owner. The annual vehicle tax is also calculated as part of the transfer process.

UC11 – Calculate Vehicle Tax

The system calculates the annual tax based on the vehicle type and its characteristics. The calculation rules are:

- **Motorcycles:** 10% of engine displacement.
- **Petrol cars:** €1.40 per gram of CO2 emissions.
- **Diesel cars:** €1.80 per gram of CO2 emissions.
- **Hybrid cars:** €1.20 per gram of CO2 emissions.

This functionality is reused by the vehicle assignment, ownership transfer, and annual tax report use cases.

UC12 – Generate Annual Tax Report

Once a year, the City Administrator requests the annual tax report. The system retrieves the registered people and their vehicles, calculates the annual tax for each vehicle, and determines the total tax amount for each owner.

The report contains each owner's information, their vehicles, the tax associated with each vehicle, and the total annual tax payable by that owner.

People without vehicles have no vehicle tax to pay, while vehicles without an owner are not included in any owner's tax total.